

# Avengers Assembly: We Didn’t Know What You’d Like, So We Put Everything In

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## ABSTRACT

We present the non-standard cosmology suite of the cURAMSES adaptive mesh refinement (AMR) code (cURAMSES III), which assembles, within a single code base, essentially every popular extension of  $\Lambda$ CDM: (i) **self-interacting dark matter** (SIDM) with constant, velocity-dependent, anisotropic and inelastic (two-state) cross-sections, implemented as cell-based Monte Carlo scattering with machine-precision momentum and energy conservation; (ii) a **dark energy and non-GR gravity** battery comprising CPL dark energy with clustering perturbations, massive-neutrino linear response, early dark energy, field-level quintessence and purely kinetic k-essence, coupled quintessence, parametrized quasi-static Horndeski gravity, Hu–Sawicki  $f(R)$ , nDGP, symmetron, the environmentally damped dilaton, the parameter-free tracker cubic Galileon, and QUMOND — all built on a unified Poisson-solver framework in which model corrections compose multiplicatively, with an FFT-accelerated Newton stage for the nonlinear scalar fields; (iii) **fuzzy dark matter** (FDM), a hybrid Schrödinger–Poisson integrator with a spectral kinetic operator on the base level and finite differences on refined levels; and (iv) **evaporating primordial black holes** (PBH), a table-driven global mass-function scheme with optional local heating. Every dark-energy and gravity module has been verified at the equation level against the literature, at the solver level against analytic linear responses (to  $10^{-5}$  or better), and end-to-end with a controlled  $64^3$  simulation suite compared against an initial-conditions-consistent linear forward model, agreeing to 2–5 per cent with residual signs explained by screening. We document the algorithms, unit conventions, namelist interfaces and the verification methodology, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

**Key words:** methods: numerical – cosmology: simulations – dark energy – dark matter – gravitation – neutrinos

## 1 INTRODUCTION

The  $\Lambda$ CDM concordance model provides an excellent fit to a wide range of cosmological observations (Planck Collaboration 2020), yet the physical nature of both dark components remains unknown. On the dark-energy side, the cosmological constant may be replaced by a dynamical field with an evolving equation of state (Chevallier & Polarski 2001; Linder 2003), by an early dark energy component (Poulin et al. 2019), or the acceleration may signal a breakdown of General Relativity on cosmological scales, as in  $f(R)$  gravity (Hu & Sawicki 2007), braneworld models (Schmidt 2009), Galileon (Barreira et al. 2013), symmetron and dilaton theories (Hinterbichler & Khoury 2010; Brax et al. 2010), or MOND (Milgrom 1983). On the dark-matter side, small-scale tensions motivate self-interacting dark matter (Spergel & Steinhardt 2000; Tulin & Yu 2018), ultralight (fuzzy) dark matter (Hui et al. 2017; Schive et al. 2014), and primordial black holes (Carr & Kühnel 2021), while neutrino oscillations guarantee a small but non-zero hot component (Lesgourgues & Pastor 2006).

Stage-IV surveys will constrain all of these at the per-cent level, and their nonlinear signatures can only be predicted by simulations. Historically each extension has lived in its own specialised code; comparing models then convolves physical differences with code systematics. This paper describes the opposite approach: one AMR code, cURAMSES (Kim 2026), carrying *all* of the above extensions behind uniform namelist switches, with shared infrastructure (domain decomposition, Morton-key octree, multigrid/CG/FFT Poisson solvers, OpenMP+MPI parallelism) and a common verification methodology. Hence the title<sup>1</sup>.

The paper is organised in four physics sections, one per family: Section 2 describes self-interacting dark matter; Section 3 the dark energy and non-GR gravity battery, including the unified Poisson framework, the nonlinear scalar-field solvers and their verification; Section 4 fuzzy dark matter; and Section 5 evaporating primordial black holes. Section 6 concludes. Throughout, code quantities use the supercomoving variables of Martel & Shapiro (1998):  $\tilde{x} = x_{\text{com}}/(aB)$

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<sup>1</sup> From the Korean “*mwol joahalji mollaseo da neoesseo*” — we didn’t know what you’d like, so we put everything in.

(with  $B$  the box size),  $\tilde{\rho} = \rho a^3/(\rho_c \Omega_m)$  (so  $\tilde{\rho} = 1$ ),  $\tilde{\phi} = a^2 \phi/(BH_0)^2$  and  $dt = H_0 dt/a^2$ .

## 2 SELF-INTERACTING DARK MATTER

### 2.1 Physical motivation

SIDM models introduce a non-zero scattering cross-section  $\sigma/m$  that allows dark matter particles to exchange momentum and energy, potentially resolving the core-cusp, missing satellites and too-big-to-fail problems (Spergel & Steinhardt 2000; Tulin & Yu 2018). Astrophysical constraints span dwarf galaxies ( $v \sim 30\text{--}100 \text{ km s}^{-1}$ ) to clusters ( $v \sim 10^3\text{--}3 \times 10^3 \text{ km s}^{-1}$ ); a constant  $\sigma/m \sim 1 \text{ cm}^2 \text{ g}^{-1}$  fits dwarf scales but may violate cluster bounds, motivating velocity-dependent cross-sections from light-mediator models (Feng et al. 2009; Tulin et al. 2013; Kaplinghat et al. 2016).

### 2.2 Monte Carlo scattering algorithm

We implement cell-based Monte Carlo scattering following the standard mesh/SPH approach (Rocha et al. 2013; Robertson et al. 2017). In every AMR leaf cell, dark matter particles are collected from the particle linked list, shuffled with the Fisher–Yates algorithm, and paired sequentially. For a pair with relative velocity  $v_{\text{rel}}$  the scattering probability per step is

$$P = \frac{\sigma/m(v) \bar{m} v_{\text{rel}} \Delta t}{V_{\text{cell}}} (N_{\text{dm}} - 1), \quad (1)$$

with  $\bar{m}$  the mean particle mass,  $V_{\text{cell}}$  the physical cell volume and the factor  $(N_{\text{dm}} - 1)$  accounting for all possible partners. A scatter occurs when a uniform deviate falls below  $P$ ; post-scattering velocities are constructed in the centre-of-mass frame,

$$\mathbf{v}'_i = \mathbf{v}_{\text{cm}} + \frac{m_j}{m_i + m_j} |\mathbf{v}'_{\text{rel}}| \hat{\mathbf{n}}, \quad \mathbf{v}'_j = \mathbf{v}_{\text{cm}} - \frac{m_i}{m_i + m_j} |\mathbf{v}'_{\text{rel}}| \hat{\mathbf{n}}, \quad (2)$$

which conserves momentum by construction and kinetic energy exactly for elastic scattering.

### 2.3 Cross-sections and angular distributions

Three velocity dependences are supported (`sidm.type`): constant  $\sigma/m = \sigma_0$ ; Yukawa (Born-regime)  $\sigma(v)/m = \sigma_0[1 + (v/v_0)^2]^{-2}$ ; and a generic power law  $\sigma(v)/m = \sigma_0(v/v_0)^n$ . The angular distribution is isotropic by default, or forward-peaked Rutherford-like,  $d\sigma/d\Omega \propto (1 - \cos\theta + 2\varepsilon)^{-2}$  (Kahlhoefer et al. 2014), sampled by CDF inversion.

### 2.4 Inelastic (two-state) SIDM

A two-state model (Tucker-Smith & Weiner 2001; Vogelsberger et al. 2019) adds a mass splitting  $\delta$ : ground–ground pairs can up-scatter ( $g+g \rightarrow e+e$ , allowed when  $\frac{1}{2}\mu v_{\text{rel}}^2 > \delta$ ), excited pairs down-scatter exothermically, and mixed pairs scatter elastically; the relative speed is adjusted to conserve total (kinetic + internal) energy. An initial excited fraction is assigned at start-up in a restart-safe manner. A dissipative channel (`sidm.fdis`) and a DM–baryon drag term (`sidm.baryon`) are also available.

## 2.5 Timestep, threading and diagnostics

The maximum per-level probability  $P_{\text{max}}$  is fed back into the timestep as  $\Delta t_{\text{new}} \leq (C_{\text{SIDM}}/P_{\text{max}}) \Delta t_{\text{old}}$  (default  $C_{\text{SIDM}} = 0.1$ ). The sweep is OpenMP-parallel over grids with per-thread decorrelated RNG streams and deterministic reseeding, and cumulative momentum/energy residuals are reduced across MPI ranks each step, providing runtime verification of the conservation properties. All parameters live in `&SIDM_PARAMS`.

## 3 DARK ENERGY AND NON-GR GRAVITY

This section describes the largest part of the suite: eleven dark-energy and modified-gravity models sharing one solver infrastructure. Table 1 summarises the models, their parameters and their verification status.

### 3.1 Unified Poisson-solver framework

In supercomoving units the code solves  $\tilde{\nabla}^2 \tilde{\phi} = \frac{3}{2} \Omega_m a (\tilde{\rho} - 1)$ . All smooth-dark-sector corrections enter this equation as multiplicative factors on the source. A single function, `cosmo_poisson_fourpi()`, provides the prefactor consistently to the multigrid, conjugate-gradient and coarse-level solvers, and the FFT solvers apply the same corrections per Fourier mode in the Green’s function, so that scale-dependent responses cost nothing extra. The factors compose:

$$\frac{3}{2} \Omega_{cb} a \times \underbrace{\left[1 + \frac{\Omega_\nu}{\Omega_{cb}} R_\nu(k, a)\right]}_{\text{neutrinos}} \underbrace{\left[1 + \frac{\Omega_{\text{DE}}(a)}{\Omega_{cb}} R_{\text{DE}}(k, a)\right]}_{\text{DE clustering}} \underbrace{\mu(a, k)}_{\text{Horndeski coupled DE}} \underbrace{e^{-\beta(\varphi - \varphi_0)}}_{\text{DE}}. \quad (3)$$

### 3.2 CPL dark energy with clustering perturbations

The background uses the exact CPL integral  $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)} e^{-3w_a(1-a)}$  in the supercomoving Friedmann factors, the growth factor and the initial velocity scaling. Perturbations are supported in two modes:

(i) *Table mode*. A CAMB-derived table of the effective ratio  $R_{\text{DE}}(k, a) = \delta_{\text{DE}}/\delta_m$  is measured with the Weyl-potential method (the ratio of potential per unit matter contrast between a clustering-DE and a smooth-DE universe), which yields the Newtonian-gauge response appropriate for the Poisson source; synchronous-gauge  $\delta_{\text{DE}}$  would over-correct by an order of magnitude.

(ii) *Quasi-static fallback*. When no table is supplied and  $c_s^2 > 0$ , the code applies the Sapone & Kunz (2009) closure

$$\frac{\delta_{\text{DE}}}{\delta_m} = \frac{1 + w}{(1 - 3w) + \frac{2c_s^2 k^2}{3a^2 H^2}}, \quad \text{i.e. d.f.} = 1 + \frac{\alpha}{k^2 + \kappa^2}, \quad (4)$$

with  $\kappa^2 = \frac{3}{2}(1 - 3w)a^2 E^2(L/\lambda_H)^2/c_s^2$  and  $\alpha = (\Omega_\Lambda f_{\text{DE}} a^3/\Omega_m)(1 + w) \frac{3}{2} a^2 E^2(L/\lambda_H)^2/c_s^2$ . Correct limits: a  $k$ -independent boost  $1 + (\rho_{\text{DE}}/\rho_m)(1 + w)/(1 - 3w)$  for  $c_s^2 \rightarrow 0$ , unity below the sound horizon, and bit-wise unity for  $\Lambda$ . Validated against CAMB (same Weyl method) to  $\leq 0.1$  per cent for  $(w_0, w_a) = (-0.9, 0.1)$  and  $\leq 0.8$  per cent for  $w_0 = -0.8$  at  $z \geq 0.5$  ( $\sim 4$  per cent at  $z = 0$ , where the quasi-static approximation degrades — use the table mode for precision).

**Table 1.** The dark energy and gravity battery. “Linear response” lists the analytically validated unscreened limit; the last column the accuracy achieved in dedicated unit tests of the discretised equations (see Section 3.10).

| Model                         | Switch / namelist                  | Background                | Linear response                                             | Unit-test accuracy                                |
|-------------------------------|------------------------------------|---------------------------|-------------------------------------------------------------|---------------------------------------------------|
| CPL DE + perturbations        | <code>w0,wa,cs2_de,de_table</code> | $w = w_0 + w_a(1 - a)$    | Sapone–Kunz closure                                         | $\leq 0.1\text{--}0.8\%$ vs CAMB ( $z \geq 0.5$ ) |
| Massive $\nu$ linear response | <code>use_neutrino</code>          | —                         | $1 + (\Omega_\nu/\Omega_{cb})R_\nu(k, a)$                   | table-exact                                       |
| Early dark energy             | <code>use_ede</code>               | Poulin+19 fluid           | smooth                                                      | analytic dilution exact                           |
| Quintessence                  | <code>use_quintessence</code>      | Klein–Gordon (shooting)   | smooth ( $c_s^2 = 1$ )                                      | $w$ -consistency $2 \times 10^{-5}$               |
| k-essence                     | <code>use_kessence</code>          | $P(X) = -X + X^2$ (exact) | Helmholtz with $c_s^2(a)$                                   | analytic limits exact                             |
| Coupled quintessence          | <code>use_coupled_de</code>        | KG + mass evolution       | $G_{\text{eff}} = G(1 + 2\beta^2)$                          | isolation runs $\leq 1\%$                         |
| Horndeski $\mu(a, k)$         | <code>use_horndeski</code>         | $\Lambda$ CDM/CPL         | $\mu = 1 + \mu_0\Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$ | exact by construction                             |
| $f(R)$ Hu–Sawicki             | <code>use_fR</code>                | $\Lambda$ CDM             | $\mu = 1 + \frac{1}{3} \frac{k^2}{k^2 + m^2 a^2}$           | $6 \times 10^{-7}$                                |
| nDGP                          | <code>use_nDGP</code>              | $\Lambda$ CDM             | $G_{\text{eff}}/G = 1 + 1/(3\beta)$                         | force field $1.5 \times 10^{-6}$                  |
| Symmetron                     | <code>use_symmetron</code>         | $\Lambda$ CDM             | $F_5/F_N = 2\beta^2\chi^2$                                  | exact (Brax+12 eq. 68 match)                      |
| Dilaton (Brax+12)             | <code>use_dilaton</code>           | $\Lambda$ CDM             | $2\beta(a)^2 k^2/(k^2 + m^2 a^2)$                           | 5 digits                                          |
| Cubic Galileon (tracker)      | <code>use_galileon</code>          | tracker $E^2(a)$          | $G_{\text{eff}}/G(a=1) = 1.844$                             | 5 digits                                          |
| QUMOND                        | <code>use_mond</code>              | —                         | $\nu( g_N /a_0)$                                            | $\mu\nu$ conjugacy $2 \times 10^{-16}$            |

at low redshift). An earlier release used a closure with the opposite sign (a 15–29 per cent source suppression); results obtained with it should be discarded.

### 3.3 Massive neutrino linear response

Neutrino perturbations are added without neutrino particles:  $\delta_\nu(k, a) = R_\nu(k, a) \delta_{cb}(k, a)$  with  $R_\nu$  precomputed from CAMB (Lewis et al. 2000), entering the Poisson source as in Section 3.1. This avoids neutrino shot noise entirely and is exact at linear order (cf. Ali-Haïmoud & Bird 2013).

### 3.4 Early dark energy

The Poulin et al. (2019) fluid form  $\rho_{\text{EDE}}(a) = \omega_{\text{ede}} \rho_{\text{std}}(a_c) 2/[1 + (a/a_c)^{3(1+w_{\text{ede}})}]$  is added to the Friedmann factors. The implementation reproduces the analytic dilution slope  $-3(1+w)$  and the analytic peak  $f_{\text{EDE}}^{\text{max}} = 1.14\omega_{\text{ede}}$  at  $a = 3^{1/4}a_c$  exactly. Two conventions matter when mapping CMB constraints:  $\omega_{\text{ede}}$  is the fraction at  $a_c$  (not the peak), and the background carries no radiation, so at  $z_c \sim 3000$  the fraction differs from Planck-style  $f_{\text{EDE}}$  by  $\sim 2\times$ .

### 3.5 Scalar-field dark energy: quintessence and k-essence

A dedicated module tabulates scalar-field backgrounds on a  $\ln a$  grid ( $a = 10^{-6}\text{--}1.55$ ) with lazy, self-healing initialisation (the tables rebuild automatically if the cosmology is overridden by IC headers).

*Quintessence* (Ratra & Peebles 1988): the Klein–Gordon and Friedmann equations are integrated with RK4 for Ratra–Peebles ( $V \propto \varphi^{-\alpha}$ ) or exponential potentials, the amplitude fixed by shooting to  $\Omega_\varphi(a=1) = \Omega_\Lambda$ . The tabulated  $w(a)$  obeys  $d \ln f_{\text{DE}}/d \ln a = -3(1+w)$  to  $2 \times 10^{-5}$ , and the RP tracker attractor  $w = -2/(\alpha + 2)$  is recovered in the matter era.

*k-essence* (Armendariz-Picon et al. 2001): for the purely kinetic  $P(X) = -X + X^2$  the shift-symmetry integral  $P_{,X}\sqrt{2X}a^3 = \text{const}$  gives  $X(a)$  algebraically;  $w \rightarrow 1/3$  and  $c_s^2 \rightarrow 1/3$  at early times (a dark-radiation-like component) and  $w \rightarrow -1$  today. The model’s  $c_s^2(a)$  feeds the Helmholtz closure of equation (4) automatically.

### 3.6 Coupled quintessence

Conformally coupled dark matter,  $m_{\text{dm}} \propto e^{-\beta\varphi/M_{\text{Pl}}}$  (Amen-dola 2000), adds four effects, all implemented and individually verified: (i) the modified expansion, including the DM mass-evolution factor in the *matter* term of the Friedmann integrals (its omission is detectable: an isolation run reproduced the inconsistent- $H$  growth prediction to 0.2 per cent); (ii) the same factor multiplying the Poisson source; (iii) the momentum-conservation velocity term  $+\beta\dot{\varphi}\mathbf{v}$ , split across the two half-kicks of the KDK integrator; and (iv) the fifth-force enhancement  $G_{\text{eff}} = G(1 + 2\beta^2)$  between matter particles.

### 3.7 Parametrized Horndeski gravity

The quasi-static, scale-independent limit of Horndeski gravity is implemented as  $\mu(a) = 1 + \mu_0\Omega_{\text{DE}}(a)/\Omega_{\text{DE},0}$  multiplying the Poisson source, with an optional Compton-mass  $k$ -dependence  $\mu(a, k) = 1 + [\mu(a) - 1]k^2/(k^2 + a^2m^2)$  applied exactly in the FFT solver paths (the multigrid path uses the  $k \rightarrow \infty$  limit).

### 3.8 Screened modified gravity: the scalar-field solvers

Five models solve a nonlinear elliptic equation for an extra scalar degree of freedom on the AMR hierarchy, with the fifth force added to the particle acceleration.

#### 3.8.1 $f(R)$ Hu–Sawicki

The quasi-static equation in code units is

$$\tilde{\nabla}^2 f_R = \frac{a^2}{3} \left( \frac{H_0 B}{c} \right)^2 [\tilde{R}(f_R) - \tilde{R}] - \Omega_m \left( \frac{H_0 B}{c} \right)^2 \frac{\tilde{\rho} - 1}{a}, \quad (5)$$

with the Hu–Sawicki inversion  $\tilde{R} = \tilde{R}_0(f_{R0}/f_R)^{1/(n+1)}$  and force  $F_5 = +\frac{a^2}{2}(c/H_0 B)^2 \tilde{\nabla}^2 f_R$  (Hu & Sawicki 2007; Oyaizu 2008; Li et al. 2012). The linearized mass term is a Yukawa (the Jacobian is negative definite) and the chameleon limit  $\tilde{R} \rightarrow \tilde{R} + 3\Omega_m \delta/a^3$  is exact. The discretised linear response reproduces  $\mu(k) - 1 = \frac{1}{3}k^2/(k^2 + m^2 a^2)$  to  $6 \times 10^{-7}$  across  $a \in [0.25, 1]$ ,  $f_{R0} \in \{-10^{-5}, -10^{-6}\}$ ,  $n \in \{1, 2\}$ .

### 3.8.2 nDGP

The brane-bending mode obeys the Vainshtein equation (Schmidt 2009; Li et al. 2013); in code units

$$\tilde{\nabla}^2 \varphi + \frac{1}{12 \Omega_{rc} \beta(a) a^4} [(\tilde{\nabla}^2 \varphi)^2 - (\tilde{\nabla}_i \tilde{\nabla}_j \varphi)^2] = \frac{\Omega_m a}{\beta(a)} (\tilde{\rho} - 1), \quad (6)$$

with  $\beta = 1 + 2Hr_c(1 + \dot{H}/3H^2)$  and  $F_5 = -\frac{1}{2}\tilde{\nabla}\varphi$ , so that the unscreened limit is  $G_{\text{eff}}/G = 1 + 1/(3\beta)$  (Winther et al. 2015).

### 3.8.3 Symmetron

The dimensionless field  $\chi = \varphi/\varphi_*$  obeys  $\tilde{\nabla}^2 \chi = \frac{a^2}{2\tilde{\chi}^2} [\tilde{\rho}(a_{\text{ssb}}/a)^3 - 1]\chi + \frac{a^2}{2\tilde{\chi}^2} \chi^3$  (Hinterbichler & Khourey 2010; Davis et al. 2012), identical to eq. (68) of Brax et al. (2012), with force  $F_5 = -6\Omega_m \beta^2 (\lambda/B)^2 (a^2/a_{\text{ssb}}^3) \chi \tilde{\nabla}\chi$ , whose unscreened limit is the canonical  $F_5/F_N = 2\beta^2 \tilde{\chi}^2$ . Because  $\chi = 0$  is an exact fixed point of the homogeneous equation, the broken-phase VEV  $\bar{\chi}(a) = \sqrt{1 - (a_{\text{ssb}}/a)^3}$  is seeded into uninitialised cells at every solve (this also covers restarts and newly refined cells).

### 3.8.4 Environmentally damped dilaton

The original  $r = 3/2$  dilaton of Brax et al. (2010), with  $A(\varphi) = 1 + \frac{A_2}{2}\varphi^2/M_{\text{Pl}}^2$ ,  $m^2(a) = 3A_2 H^2(a)$  and  $\beta(a) = \beta_0 a^{3\Omega_m}$  (Brax et al. 2012), is solved in  $\chi = \varphi/M_{\text{Pl}}$ :

$$\tilde{\nabla}^2 \chi = \frac{3\Omega_m A_2 B_2}{a} (\tilde{\rho} \chi - \bar{\chi}) + a^2 B_2 [v_\varphi(\chi) - v_\varphi(\bar{\chi})], \quad (7)$$

with  $v_\varphi(\chi) = -3\Omega_m \beta_0 (A_2 \chi / \beta_0)^{1-3/s}$ ,  $s = 3\Omega_m$ ,  $B_2 = (H_0 B/c)^2$  and  $F_5 = -(c/H_0 B)^2 a^2 A_2 \chi \tilde{\nabla}\chi$ . The exponent  $1 - 3/s < 0$  makes the Newton Jacobian negative definite. The background is an exact fixed point of the discrete operator, the linear response matches  $2\beta(a)^2 k^2/(k^2 + m^2 a^2)$  to five digits, and a  $\delta = 10^5$  top-hat drives  $\chi \rightarrow 0.03\bar{\chi}$  (Damour–Polyakov screening). User parameters are simply  $\beta_0$  and the present force range  $\lambda = 2998 \xi h^{-1} \text{Mpc}$  (then  $A_2 = 1/3\xi^2$ ).

### 3.8.5 Tracker cubic Galileon

On the tracker,  $H\dot{\varphi} = \xi H_0^2 M_{\text{Pl}}$  with  $c_2 = -6c_3 \xi$ , the model is parameter-free:  $\xi = \sqrt{6(1 - \Omega_m)}$ ,  $E^2(a) = \frac{1}{2}[\Omega_m a^{-3} + \sqrt{\Omega_m^2 a^{-6} + 4(1 - \Omega_m)}]$  (wired through the background dispatch, so the modified expansion and growth are automatic;  $w_0 \simeq -1.18$ ), and the perturbation coefficients of Barreira et al. (2013) reduce to  $\beta_1 = \frac{2}{3}[2\dot{H}/H^2 - 1 + (1 - \Omega_m)/E^4]$ ,  $\beta_2 = 2E^2 \beta_1 / \xi^2$ . The field equation is the nDGP template with coefficient  $1/(3\beta_1 a^4)$  and source  $\Omega_m a \delta / \beta_2$ ; the Poisson back-reaction term integrates to a potential proportional to the field, so the entire fifth force is one gradient,  $F_5 = +\frac{\xi}{6E^2} \tilde{\nabla}u$ . The unscreened response matches the analytic  $-\xi/(9\beta_2 E^2)$  to five digits at  $a = 0.25$ –1:  $G_{\text{eff}}/G(a=1) = 1.844$ , decaying as  $1/E^4$  into the past. Since  $G_{\text{eff}}/G - 1 < 10^{-3}$  for  $z \gtrsim 2.5$ , the solver is skipped at early times.

### 3.8.6 QUMOND

The quasi-linear formulation of MOND (Milgrom 2010) defines a curl-free field via  $\nabla^2 \Phi = \nabla \cdot [\nu(|\nabla \Phi_N|/a_0) \nabla \Phi_N]$ . Both

an algebraic force-correction mode and the full phantom-density mode (compute  $\rho_{\text{ph}} = \nabla \cdot [(\nu - 1) \nabla \Phi_N] / 4\pi G$ , re-solve Poisson) are implemented, with the simple and standard interpolation pairs (verified mutually conjugate to  $2 \times 10^{-16}$ ), an external-field option, and  $a_0$  converted to code units through the exact supercomoving acceleration factor (so the MOND scale is correctly constant in proper units at all redshifts). An AQUAL fixed-point mode is also available (cf. Lügghausen et al. 2015; Candlish et al. 2015).

## 3.9 Numerical solution of the scalar equations

All five scalar solvers share one machinery: nonlinear Newton–Gauss–Seidel relaxation with true three-dimensional red–black colouring (cell parity  $i+j+k$ ), Neumann closure of the stencil at coarse–fine boundaries, one-sided force gradients at level edges, and MPI ghost exchange each sweep. On the fully refined domain level the long-wavelength modes — which plain relaxation cannot converge in  $\mathcal{O}(10)$  sweeps — are solved spectrally: a Newton–Helmholtz stage ( $\tilde{\nabla}^2 - \bar{m}^2$ )  $\delta u = -\mathcal{R}$  with the level-averaged Jacobian mass for  $f(R)$ /symmetron/dilaton, and a Winther et al. (2015)-style operator splitting for the Vainshtein-type models (solve the cell-local quadratic for  $\tilde{\nabla}^2 \varphi$ , then invert the Laplacian by FFT). In a controlled test the operator-split stage reached a lower residual in five iterations than  $2 \times 10^4$  damped relaxation sweeps. Two MPI subtleties deserve mention as a warning to implementers: every rank must enter the per-level ALLREDUCE of the convergence test and the final force ghost-exchange *even when it owns no grids on that level*; early returns on empty ranks deadlock or abort the run at the first sparsely refined level.

## 3.10 Verification

Verification proceeded on three levels.

*Equation level.* Every model was re-derived from its original reference (unit conversions included) and the code compared term by term; this stage found and fixed, among others, a wrong-sign DE-perturbation closure, a tachyonic (non-convergent)  $f(R)$  operator, a double-counted nDGP coupling, a symmetron force normalisation missing the box-unit conversion, and a zero-pinned symmetron field.

*Solver level.* The discretised operators were exercised in isolation against analytic linear responses; the accuracies are listed in Table 1.

*End-to-end level.* A controlled suite of  $64^3$ ,  $100 h^{-1} \text{Mpc}$  dark-matter-only simulations was run from shared Zel’dovich initial conditions ( $z = 49$ , CAMB spectrum) for  $\Lambda \text{CDM}$ , quintessence, k-essence, Horndeski, coupled DE (plus per-effect isolation runs), the tracker Galileon and the dilaton. The measured  $P_{\text{model}}/P_{\Lambda \text{CDM}}$  in the band  $k = 0.05$ – $0.12 h \text{Mpc}^{-1}$  at  $z \simeq 0$  was compared against an *initial-conditions-consistent* linear forward model, which (i) rescales the initial displacements by each model’s own velocity conversion factor — RAMSES converts the shared velocity files with the model’s  $v_{\text{fact}}$ , an effect reaching  $(1.077)^2$  for k-essence and  $(1.109)^2$  for coupled DE — and (ii) propagates the exact  $(\delta, \theta)$  initial vector through each model’s linear system. Results:

All models agree to 2–5 per cent; the Galileon and dilaton deficits carry the expected sign, since Vainshtein and



**Table 2.** End-to-end suite: simulated vs. predicted  $P/P_{\Lambda\text{CDM}}$  ( $k = 0.05\text{--}0.12\ h/\text{Mpc}$ ,  $z \simeq 0$ ).

| Model                                    | Simulation | Forward model | Ratio |
|------------------------------------------|------------|---------------|-------|
| Quintessence (RP $\alpha=1$ )            | 0.832      | 0.817         | 1.018 |
| k-essence ( $X_0/M^4=0.5005$ )           | 0.875      | 0.881         | 0.993 |
| Horndeski ( $\mu_0=0.2$ )                | 1.036      | 1.052         | 0.985 |
| Coupled DE ( $\beta=0.1$ )               | 0.966      | 1.005         | 0.961 |
| Galileon (tracker)                       | 1.148      | 1.208         | 0.950 |
| Dilaton ( $\beta_0=0.5$ , $\lambda=20$ ) | 1.039      | 1.071         | 0.970 |

Damour–Polyakov screening remove part of the fifth-force boost already at quasi-linear wavenumbers (cf. Barreira et al. 2013). Each coupled-DE ingredient (mass evolution, friction,  $G_{\text{eff}}$ ) was additionally isolated with toggle runs and verified at the  $\sim 1$  per cent level. The forward model, the IC generator and all unit-test drivers ship with the code (`patch/cuRamses/aux/`).

#### 4 FUZZY DARK MATTER

Ultralight scalar dark matter with  $m_a \sim 10^{-22}\text{ eV}$  (Hui et al. 2017) is evolved as a classical field  $\psi$  obeying the Schrödinger–Poisson system. The integrator uses drift–kick–drift operator splitting per AMR level: the kinetic half-steps apply  $\exp(-i\hbar k^2 \Delta t / 4a^2 m_a)$  spectrally on the base level (FFT) and an explicit finite-difference Laplacian on refined levels, while the potential kick applies  $\exp(-im_a \Phi \Delta t / \hbar)$  in real space; a fourth-order Yoshida composition is optional (`fdm_split_order=4`). Refinement follows the de Broglie wavelength (a minimum of `fdm_nrefine_dB` cells per  $\lambda_{\text{dB}}$ , with a density floor), reproducing the solitonic cores and interference fringes characteristic of FDM haloes (Schive et al. 2014). A hybrid mode (`fdm_use_hjm`) evolves a Hamilton–Jacobi–Madelung fluid on coarse levels and switches to the wave solver above `fdm_first_wave_level`, and `fdm_hybrid` allows mixed FDM + particle (stars/sinks) runs. The wavefunction is stored as  $(\psi_{\text{re}}, \psi_{\text{im}})$  on the AMR mesh with  $|\psi|^2$  the total DM density, and dedicated light-cone and output modules are provided. A detailed description and validation of the FDM solver is presented in a companion paper.

#### 5 EVAPORATING PRIMORDIAL BLACK HOLES

The PBH module models dark matter that is partly composed of Hawking-evaporating primordial black holes (Hawking 1975; Carr & Kühnel 2021). Rather than tracking per-particle masses, it uses one global mass function: all DM particles share the mixed-mass factor

$$w(a) = 1 - f_{\text{PBH}} + f_{\text{PBH}} g(a), \quad (8)$$

where  $g(a)$  is the surviving PBH mass fraction from a pre-computed evaporation table and  $f_{\text{PBH}}$  (`pbh_fraction`) the initial PBH share of the dark matter. Particle masses are rescaled exactly by table arithmetic — there is no per-particle state and no evaporation CFL constraint. The energy released between steps follows the cumulative heating kernel  $\tilde{Q}(a) = \int g g dt$ ; it is either deposited as local heat in the gas (`pbh_energy_sink='local_heat'`, optionally amplified by a

linear-response boost factor) or removed from the simulation (`'removed'`, appropriate for free-streaming decay products). Restart provenance (table checksum, normalisation epoch) is persisted so that a run cannot silently continue with a different evaporation model. The table generator supports arbitrary monochromatic or extended PBH mass functions.

#### 6 CONCLUSIONS

We have assembled, in one AMR code, the four families of non-standard cosmological physics most often invoked beyond  $\Lambda\text{CDM}$  — self-interacting dark matter, a comprehensive dark-energy and modified-gravity battery, fuzzy dark matter, and evaporating primordial black holes — behind uniform namelist switches and on shared solver infrastructure. Beyond the implementations themselves, we highlight the verification methodology: every model is checked at the equation level against its original reference, at the solver level against analytic linear responses, and end-to-end against an initial-conditions-consistent forward model, with per-effect isolation runs where a model bundles several couplings. We commend this three-level protocol to implementers of non-standard physics in any simulation code: in our experience each level catches defects invisible to the others, ranging from wrong-sign perturbation closures to factor-of-two integrator bookkeeping and MPI collective mismatches that only manifest on sparsely refined levels.

All modules recover  $\Lambda\text{CDM}$  bit-wise when disabled. The namelist interfaces, verification scripts and the simulation-suite tooling are distributed with the code.

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#### DATA AVAILABILITY

The code, namelist examples, verification drivers and the  $64^3$  test-suite configuration are available from the `CURAMSES` repository. Simulation outputs will be shared on reasonable request.

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